



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

UNIT TEST

30-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2023

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SECTION-I : (Maximum Marks: 80)

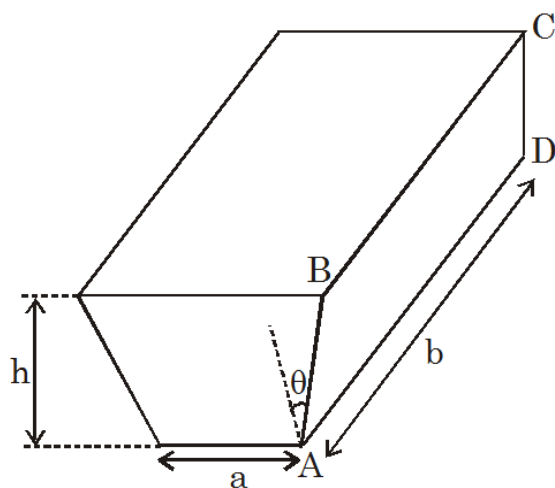
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

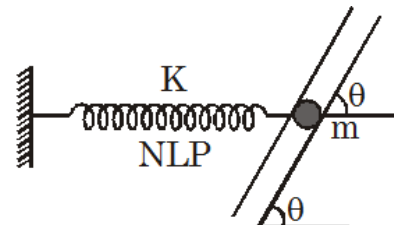
Negative Marks : -1 If wrong option is selected.

1. A tray of the shape is shown in figure. It is completely filled with a liquid of density ρ . Find the force on its rectangular wall ABCD due to liquid. The wall is inclined at an angle θ to the vertical. (Ignore atmospheric pressure)



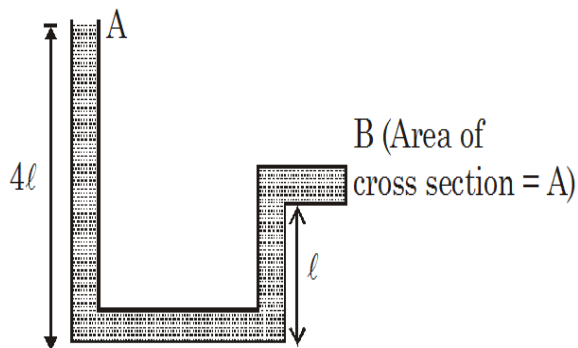
- (A) $\frac{1}{2} \rho g h^2 b$
 (B) $\frac{1}{2} \rho g h^2 b \sec \theta$
 (C) $\rho g h^2 b$
 (D) $\frac{1}{2} \rho g h^2 b \tan \theta$

2. A block of mass m is restricted to move in a smooth groove. It is connected to a spring of spring constant k . Neglecting the effect of gravity find time period of small oscillation :-



- (A) $\frac{2\pi}{\cos \theta} \sqrt{\frac{m}{k}}$
 (B) $2\pi \sqrt{\frac{m}{k}}$
 (C) $\frac{2\pi}{\sin \theta} \sqrt{\frac{m}{k}}$
 (D) $2\pi \sqrt{\frac{m \cos^2 \theta}{k}}$
3. The time period of a particle in simple harmonic motion is equal to the time between consecutive appearances of the particle at a particular point in its motion. This point is
- (A) The mean position
 (B) Between the mean position and the positive extreme
 (C) An extreme position
 (D) Between the mean position and the negative extreme.

4. A thin tube in vertical plane is shown in figure. It is filled with a liquid of density ρ and its end B is closed. Then the force exerted by the liquid on the tube at end B will be : [Neglect atmospheric pressure and the radius of the tube to be negligible in comparison to ℓ]



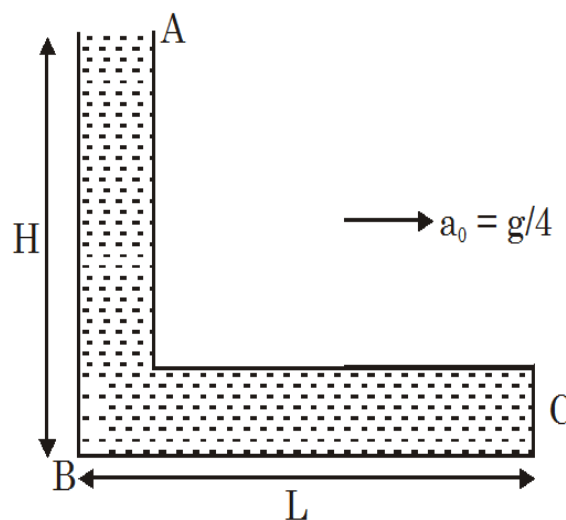
- (A) $\rho \cdot g \ell \cdot A$ (B) $3\rho g \ell A$
 (C) $\frac{\rho g \ell A}{2}$ (D) Zero
5. Potential energy associated with a particle of mass 'm' moving along x axis is given by $U = U_0[1 - \cos(bx)]$ where U_0 & b are positive constant. Find the time period of small oscillation

- (A) $2\pi\sqrt{\frac{m}{U_0 b}}$
 (B) $2\pi\sqrt{\frac{mb}{U_0}}$
 (C) $2\pi\sqrt{\frac{m}{U_0}}$
 (D) $\frac{2\pi}{b}\sqrt{\frac{m}{U_0}}$

6. The total mechanical energy of a spring-mass system in simple harmonic motion is $E = \frac{1}{2}m\omega^2 A^2$. Suppose the oscillating particle is replaced by another particle of double the mass while the amplitude A remains the same. The new mechanical energy will

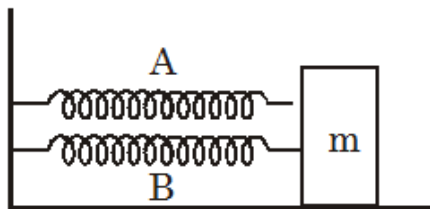
- (A) become $2E$ (B) become $E/2$
 (C) become $\sqrt{2}E$ (D) remain E

7. An L shaped thin tube contains a liquid of density ρ . It is moved horizontally with a constant acceleration $\frac{g}{4}$. Part AB of tube is vertical and BC is horizontal. End A is open to atmosphere and atmospheric pressure is P_0 . If pressure of liquid at point C (just inside the tube) is equal to P_0 then:-



- (A) $L = 4H$ (B) $L = 2H$
 (C) $L = 3H$ (D) $L = H$

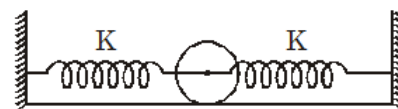
8. In the figure block lies on smooth horizontal surface and both the springs have same spring constant k . The spring B is attached with block while spring A just touches the block as the position shown and both the springs are in relaxed state. The time period of oscillation is :-



- (A) $2\pi\sqrt{\frac{m}{2k}}$
 (B) $2\pi\sqrt{\frac{3m}{2k}}$
 (C) $2\pi\sqrt{\frac{m}{k}}$
 (D) $\frac{(\sqrt{2}+1)}{\sqrt{2}}\pi\sqrt{\frac{m}{k}}$
9. A particle executes SHM about the point $x = 0$. At time $t = 0$ it has position $x = A$ and zero velocity. The angular frequency of oscillation is ω . The acceleration of the particle at $t = \frac{\pi}{4\omega}$ is :-

- (A) $-A\omega^2$ (B) $-\frac{A\omega^2}{\sqrt{2}}$
 (C) Zero (D) $-A\omega^2\sqrt{2}$

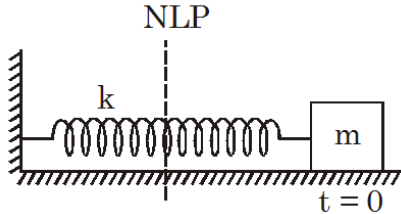
10. Two springs each of spring constant K are connected with the centre of a sphere of mass m and radius R as shown in figure. Given system lies on horizontal surface and both the springs are horizontal. Friction is sufficient to prevent relative slipping between horizontal surface and sphere. From the position where both springs are at natural length, sphere is slightly displaced by a distance ' x_0 ' and released from rest. After being released from rest find minimum time taken by sphere to move by a distance $\frac{x_0}{2}$.



- (A) $\frac{\pi}{3}\sqrt{\frac{7m}{10k}}$ (B) $\frac{\pi}{3}\sqrt{\frac{5m}{7k}}$
 (C) $\frac{\pi}{4}\sqrt{\frac{7m}{5k}}$ (D) $\frac{\pi}{6}\sqrt{\frac{5m}{7k}}$
11. A simple pendulum is suspended from the ceiling of a car moving with constant speed on a horizontal circular track of radius R . Speed of the car is given by $v = \sqrt{gR}$ where g is acceleration due to gravity and R is radius of circular turn. Find the time period of a simple pendulum of length ℓ suspended from the ceiling of the car.

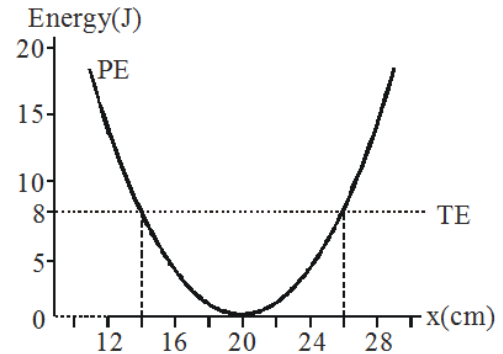
- (A) $2\pi\sqrt{\frac{\ell}{2g}}$ (B) $2\pi\sqrt{\frac{2\ell}{g}}$
 (C) $2\pi\sqrt{\frac{\ell}{g}}$ (D) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$

12. In a horizontal spring-mass system, at $t = 0$, mass 'm' is released from rest after being displaced (from NLP of spring) towards right by some distance on a frictionless surface. Find minimum time after which speed of block becomes $\frac{1}{\sqrt{2}}$ times of maximum possible speed during the motion.

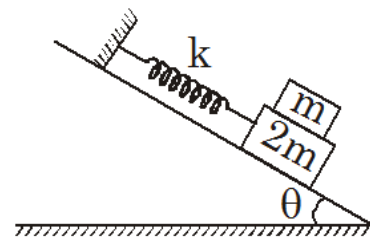


- (A) $\frac{\pi}{2} \sqrt{\frac{m}{k}}$ (B) $\frac{\pi}{4} \sqrt{\frac{m}{k}}$
 (C) $\pi \sqrt{\frac{m}{k}}$ (D) $\frac{\pi}{8} \sqrt{\frac{m}{k}}$
13. Two particles A and B are performing SHM along same straight line with same amplitude a and same angular frequency ω . At $t = 0$, A is at the left extreme and B is at a distance $a/2$ from mean position to the right of mean position and moving toward mean position. Then:
- (A) their phase difference is $\frac{5\pi}{3}$
 (B) their phase difference is $\frac{\pi}{3}$
 (C) time after which they cross each other is $\frac{2\pi}{3\omega}$
 (D) time after which they cross each other is $\frac{\pi}{3\omega}$

14. Figure shows the potential-energy diagram and the total energy line of a particle oscillating on a spring. Which of the following option is incorrect.



- (A) the amplitude of oscillation of particle is 6 cm.
 (B) the spring constant is 4.44 kN/m.
 (C) the mass of particle is 1 kg.
 (D) the maximum kinetic energy of particle is at $x = 20$ cm.
15. The coefficient of friction between block of mass m and 2m is $\mu = 2 \tan \theta$. There is no friction between block of mass 2m and inclined plane. The maximum amplitude of two block system for which there is no relative motion between both the blocks is:



- (A) $g \sin \theta \sqrt{\frac{k}{m}}$ (B) $\frac{mg \sin \theta}{k}$
 (C) $\frac{3mg \sin \theta}{k}$ (D) None of these

16. A block of ice is floating in a liquid (in beaker) of specific gravity 1.2, which is immiscible with water. When all the ice melts, the liquid level in the beaker.

(A) falls
(B) rises
(C) remains unchanged
(D) None

17. A rod of mass 'm' and length 'L' is suspended in vertical plane so as to oscillate freely about an axis normal to the plane of paper passing through 'S' (point of suspension). Find the distance of 'S' from end A of rod such that time period of small oscillation is minimum.

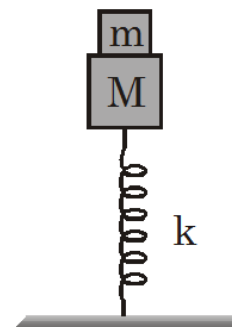


(A) $\frac{L}{2}$ (B) $\frac{L}{\sqrt{12}}$
(C) $\frac{L}{2} + \frac{L}{\sqrt{12}}$ (D) $\frac{L}{2} - \frac{L}{\sqrt{12}}$

18. A wooden cube (density of wood ' $\frac{\rho}{3}$ ') of side ' ℓ ' floats in a liquid of density ' ρ ' with its upper and lower surfaces horizontal. If the cube is pushed slightly down and released, it performs simple harmonic motion of period 'T'. Then, 'T' is equal to :-

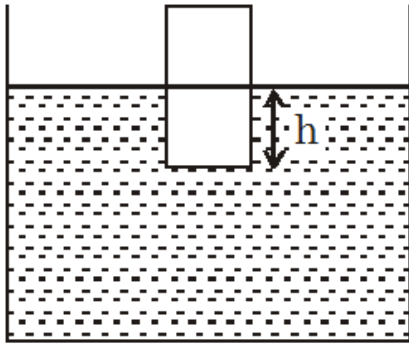
(A) $2\pi\sqrt{\frac{3\ell}{2g}}$ (B) $2\pi\sqrt{\frac{\ell}{3g}}$
(C) $2\pi\sqrt{\frac{3\ell}{g}}$ (D) $2\pi\sqrt{\frac{2\ell}{g}}$

19. A small block of mass m is kept on a bigger block of mass M which is attached to a vertical spring of spring constant k as shown in the figure. The system oscillates vertically. What can be the maximum amplitude with which the two blocks may oscillate together ?



(A) $\frac{2(M+m)g}{k}$ (B) $\frac{(m)g}{k}$
(C) $\frac{(M)g}{k}$ (D) $\frac{(M+m)g}{k}$

20. A block is floating in water in a container at rest. Length of block inside the container is h as shown in figure. Density of block is $\frac{\rho}{4}$ and density of water is ρ . Now the container starts accelerating upwards with constant acceleration $\frac{g}{4}$. The new length of block inside the water will be [consider container achieves acceleration slowly]



- (A) $\frac{h}{4}$ (B) $\frac{h}{2}$ (C) h (D) $2h$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

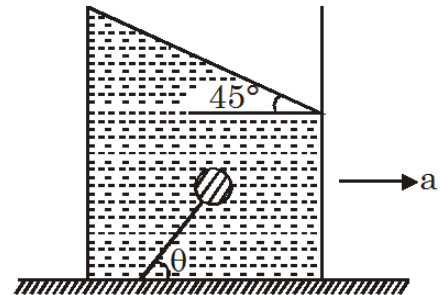
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

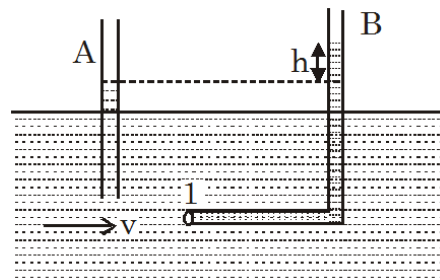
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. A solid of density $\frac{\rho}{2}$ (ρ = density of water) is held inside a water tank, tied to a string, as shown in figure. The tank is accelerated uniformly in horizontal direction such that the water surface makes an angle 45° with the horizontal. Find the angle θ (in degree) that the string makes with base of container in equilibrium.

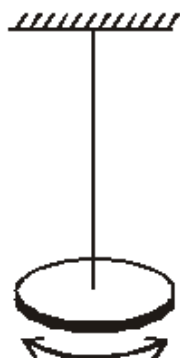


2. Liquid is flowing in a horizontal pipe. Two tubes, A & B are inserted in the pipe. Tube B is L-shaped and its cross-section at 1 is held normal to the flow direction. Liquid rises in both tubes. Height of liquid column in B is h more than that in A. Considering $h = 5$ cm and $g = 10$ m/s², find the speed of liquid flow (v) in m/sec in the pipe.

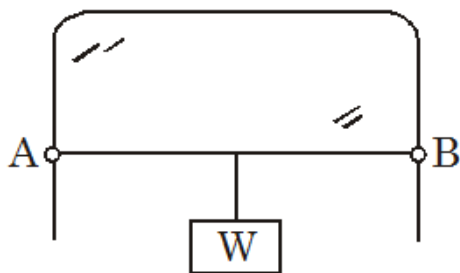


3. Equation of motion of a particle performing SHM is given by $y = \frac{10}{\pi} \sin\left[2000\pi t - \frac{\pi}{6}\right]$, where y is in cm and t is in sec. If time period of SHM is t and maximum possible velocity of the particle during SHM is $v_{\max}$ then find the value of $t \times v_{\max}$ (in SI unit).

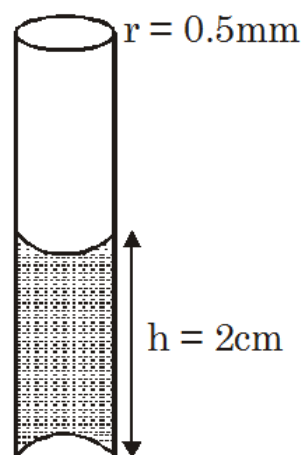
4. A uniform disc of radius 5.0 cm and mass 200g is fixed at its centre to a metal wire, the other end of which is fixed with a clamp. The hanging disc is rotated about the wire through an angle and released. If the disc makes torsional oscillations with time period 0.20 s, find the torsional constant (in S.I. units) of the wire. (Take : $\pi^2 = 10$)



5. A light wire AB of length 10 cm can slide on a vertical frame as shown in figure. There is a film of soap solution trapped between the frame and the wire. Find the load W (in g) that should be suspended from the wire to keep it in equilibrium. Neglect friction. Surface tension of soap solution = 25 dyne/cm. Take $g = 10 \text{ m/s}^2$.

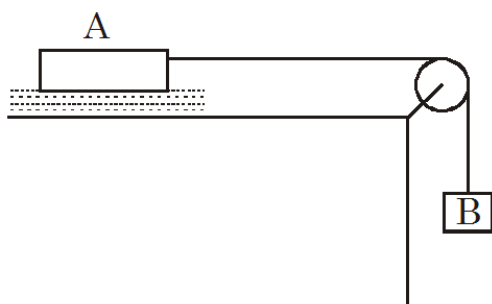


6. There is an air bubble of radius 1.0 mm in a liquid of surface tension 0.075 N/m and density 1000 kg/m^3 . The bubble is at a depth of 10 cm below the free surface. By what amount is the pressure (in N/m^2) inside the bubble greater than the atmospheric pressure ? Take : $g = 9.8 \text{ m/s}^2$.
7. A narrow capillary tube of radius 0.5 mm contains some water in it. When tube is held vertical, the water remains inside the tube as shown. Find the radius of curvature (in mm) of the lower water surface. (Surface tension $s = 0.075 \text{ N/m}$, contact angle $= 0^\circ$, $g = 10 \text{ m/s}^2$).

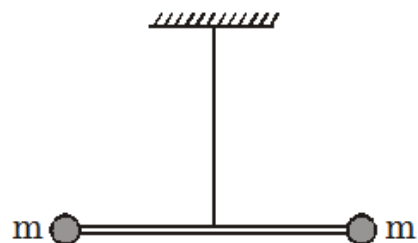


8. A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$ th of the density of material of the bob. If the bob is inside liquid all the time and its period of oscillation in this liquid is given by $4T \sqrt{\frac{1}{N}}$ then find the value of N.

9. A block A is placed on a table. There is a 3.0 mm thick layer of oil between the table and block A. Face of the block in contact with oil has an area 0.1 m^2 . The block A is connected to another block B of mass $m = 0.01 \text{ kg}$ by massless string passing over a smooth pulley. The system is released from rest. It accelerates and finally attains a constant speed of 0.085 m/s . Find coefficient of viscosity of liquid (in S.I. Units). Neglect viscous force and buoyancy force due to air on the blocks ($g = 9.8 \text{ m/s}^2$)



10. Two small balls each of mass m are connected by a uniform rod of mass m and length L . The given system is suspended from its centre by a thin wire of torsional constant k . The rod is rotated about the wire through an angle θ_0 and released. Find the velocity of each ball as the system passes through the mean position. If this velocity is given by $v = \theta_0 \sqrt{\frac{\alpha k}{7m}}$ then fill the value of α .



SECTION-I : (Maximum Marks: 80)

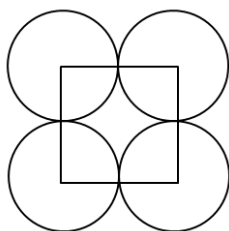
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Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

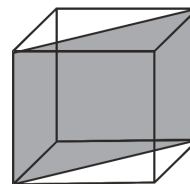
- If the value of ionic radius ratio $\left(\frac{r^+}{r^-}\right)$ is 0.52 in an ionic compound, the geometrical arrangement of ions in crystal is
(A) Tetrahedral
(B) Planar
(C) Octahedral
(D) Pyramidal
- Identical 4 spheres are taken and are arranged in a layer of square packing touching each other as shown



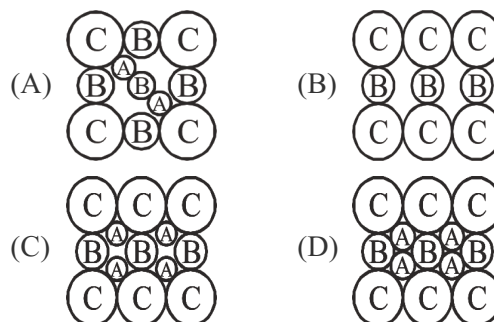
The percentage of vacant space is (in 3-D)

- $100 \left(1 - \frac{3\pi}{8}\right)$
- $100 \left(1 - \frac{\pi}{6}\right)$
- $100 - \frac{3\pi}{8}$
- $\frac{\pi}{6}$

- If $r_{\text{Na}^+} = 95 \text{ pm}$ and $r_{\text{Cl}^-} = 181 \text{ pm}$ in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions?
(A) 778 pm (B) 276 pm
(C) 195 pm (D) 390 pm
- In a hypothetical solid 'C' atoms are found to form cubical close packed lattice. 'A' atoms occupy all tetrahedral voids & 'B' atoms occupy all octahedral voids. 'A' and 'B' atoms



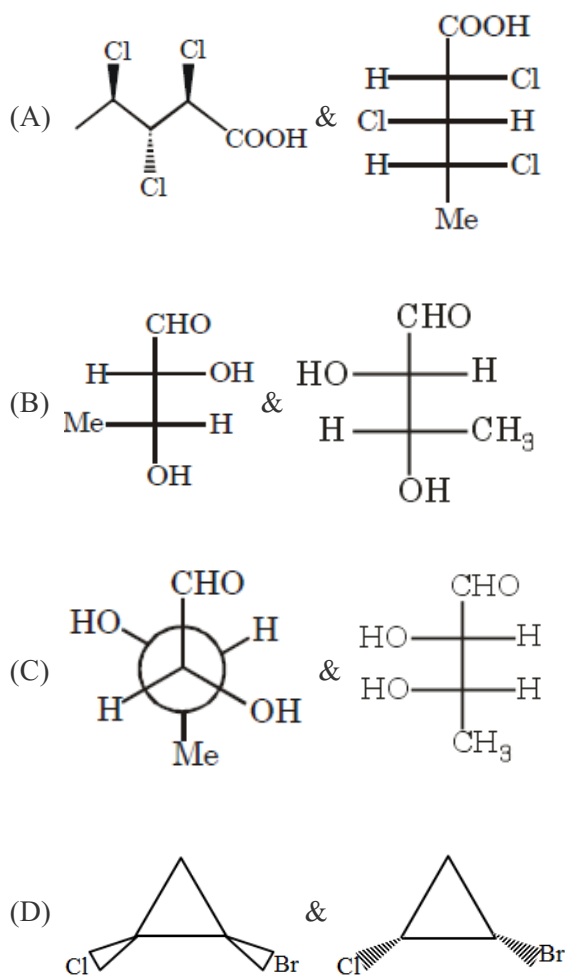
are of appropriate size i.e. there is no distortion in ccp lattice of C atoms. Now if a plane as shown in the following figure is cut, then the cross section of this plane will look like.



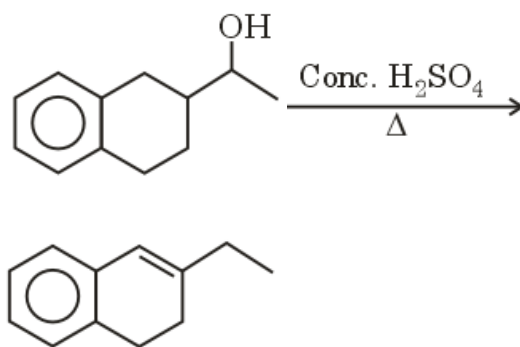
- 120 g glucose $[\text{C}_6\text{H}_{12}\text{O}_6(\text{s})]$ is dissolved to prepare 1 L aqueous solution of density 1.2g/mL. Identify incorrect option for the solution
(A) Molarity = 2/3 M
(B) % w/w = 10%
(C) % w/v = 12%
(D) Molality = 1.85 mol/kg

6. In an ionic solid atoms 'A' form hexagonal close packing crystal with atoms 'B' in $(2/3)^{\text{rd}}$ octahedral voids. Empirical formula of solid is -
 (A) AB (B) A_3B_2
 (C) A_2B (D) A_2B_3
7. $p\pi - d\pi$ bond is present in :
 (A) BF_2Cl (B) SiF_4
 (C) CO (D) BeCl_2
8. In which of the following species odd electron present in pure orbital?
 (A) NO_2 (B) ClO_2
 (C) CF_3 (D) ClO_3
9. In which of the following processes, the value of magnetic moment does not change ?
 (A) $\text{N}_2 \rightarrow \text{N}_2^-$
 (B) $\text{N}_2 \rightarrow \text{N}_2^+$
 (C) $\text{O}_2 \rightarrow \text{O}_2^+$
 (D) $\text{O}_2^+ \rightarrow \text{O}_2^-$
10. The nature of overlap in the bridge bond present in I_2Cl_6 is :
 (A) sp^3-p-sp^3
 (B) $sp^3d^2-sp^3-sp^3d^2$
 (C) $sp^3d-sp^3-sp^3d$
 (D) $sp^3d-sp^2-sp^3d$
11. Which of the following is not a correct order
 (i) $\text{AgF} < \text{AgCl} < \text{AgBr}$ (Colour intensity)
 (ii) $\text{Rb}_2\text{CO}_3 > \text{K}_2\text{CO}_3 > \text{Na}_2\text{CO}_3$ (decomposition temperature)
 (iii) $\text{BeC}_2\text{O}_4 < \text{MgC}_2\text{O}_4 < \text{CaC}_2\text{O}_4$ (Solubility)
 (iv) $\text{Li}_3\text{N} > \text{Na}_3\text{N} > \text{K}_3\text{N}$ (Thermal stability)
 (A) i, iii, iv
 (B) i, iii
 (C) only iii
 (D) iii and iv
12. Which of the following order is correct for covalent character (if the size of cations are identical in pairs)-
 (A) $\text{NaCl} < \text{CuCl}$
 (B) $\text{CaCl}_2 < \text{PdCl}_2$
 (C) $\text{RbCl} < \text{AuCl}$
 (D) All of these
13. Select **INCORRECT** order against property indicated-
 (A) $\text{AsCl}_3 < \text{SbCl}_3 < \text{BiCl}_3$ (relative stability)
 (B) $\text{ZnO} > \text{MgO} > \text{CaO} > \text{BaO}$ (Covalent character)
 (C) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$ (M.P.) Melting point
 (D) $\text{BeF}_2 > \text{MgF}_2 > \text{CaF}_2 > \text{SrF}_2 > \text{BaF}_2$ (water solubility)

14. Which of the following pair of compounds doesn't represent enantiomeric pair :

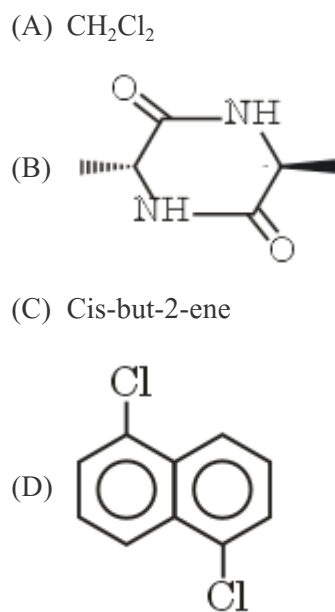


15.

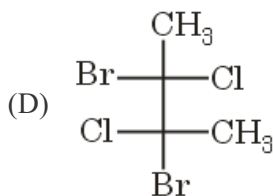
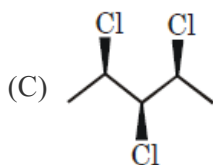
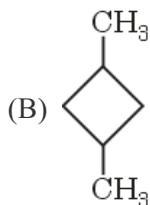
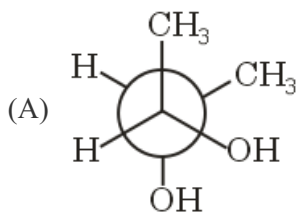


Identify total number of 1,2-shift for above reaction ?

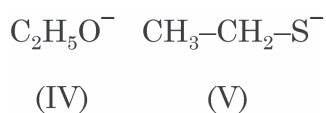
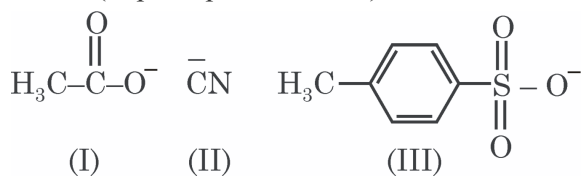
- (A) 2 (B) 3 (C) 4 (D) 5
16. Which of the following compound(s) contains plane of symmetry as well as centre of symmetry



17. Which of the following compound(s) can not exist in meso form ?

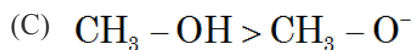
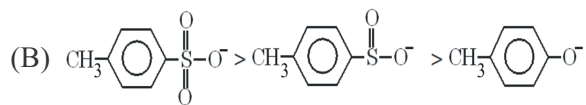
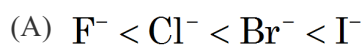


18. The order of increasing nucleophilicity of the anions (in polar protic solvent)

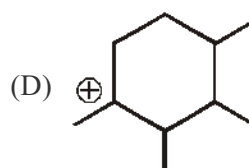
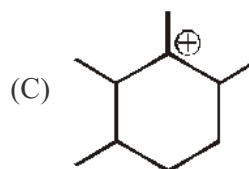
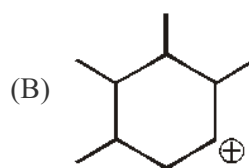
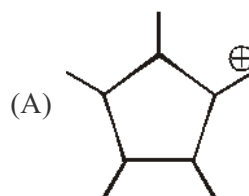
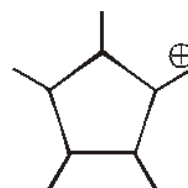


- (A) $\text{V} < \text{I} < \text{III} < \text{II} < \text{IV}$
 (B) $\text{III} < \text{I} < \text{IV} < \text{II} < \text{V}$
 (C) $\text{III} < \text{II} < \text{I} < \text{IV} < \text{V}$
 (D) $\text{II} < \text{III} < \text{I} < \text{V} < \text{IV}$

19. Which of the following order of leaving group is incorrect ?



20. The most stable carbocation after rearrangement is



SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

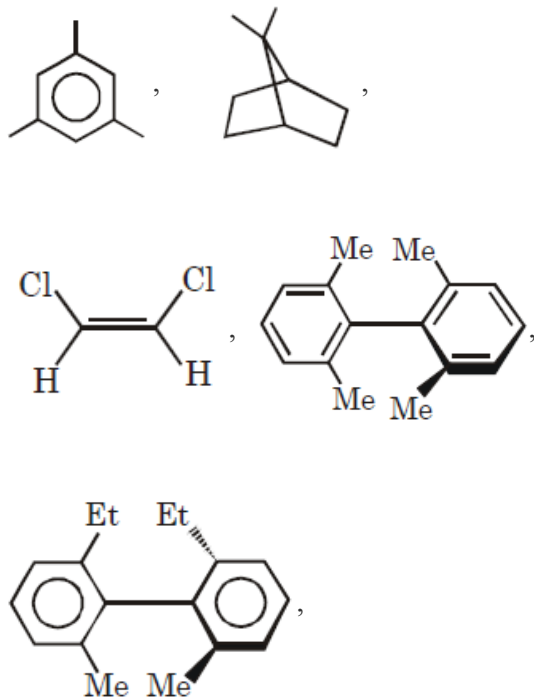
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

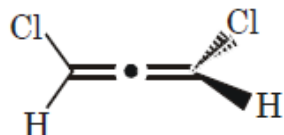
1. 2M of 100 ml Na_2SO_4 is mixed with 3M of 100 ml NaCl solution and 1M of 200 ml CaCl_2 solution. Then find the ratio of the molarity of all cation and all anions in final solution ?
2. In an ionic solid $r_{(+)} = 1.6 \text{ \AA}$ and $r_{(-)} = 1.864 \text{ \AA}$. Use the radius ratio rule to determine the edge length of the cubic unit cell in $\text{\AA}$. [$\sqrt{3} = 1.732$]

3. A sample of H_2O_2 is labelled '28 volumes' at 1 atm & 273K. Find molality of the solution if its density is 265 g/L ?
4. For any metal ^{18}M crystallising in fcc arrangement (Atomic radius = $1/\sqrt{2} \text{ \AA}$) with density is 67.5 gm/cc hence atomic mass of the metal is $[N_A = 6 \times 10^{23}]$ (in gram/mole)
5. Write the number of pairs in which size of first element or ion is higher as compare to I^{nd} out of following pairs :
(O, S), (He, Ne), (Kr, Ne), (Na, Na^+), (Cl , Cl^-), (Li, Na), (Li^+ , Na^+)
6. Find the number of element having first ionization energy lower than Be.
Mg, Ca, Sr, Ba, Li
7. How many orders are CORRECT
(a) Acidic strength $\Rightarrow \text{ClOH} < \text{BrOH} < \text{IOH}$
(b) Basic strength $\Rightarrow \text{MgO} < \text{CaO} < \text{SrO} < \text{BeO}$
(c) Electronegativity $\Rightarrow \text{I} < \text{Br} < \text{N} < \text{O} < \text{F}$
(d) Electron affinity $\Rightarrow \text{Mg} < \text{Na} < \text{Si} < \text{S} < \text{Cl}$
(e) % Ionic character $\Rightarrow \text{NaCl} < \text{KF} < \text{RbF}$

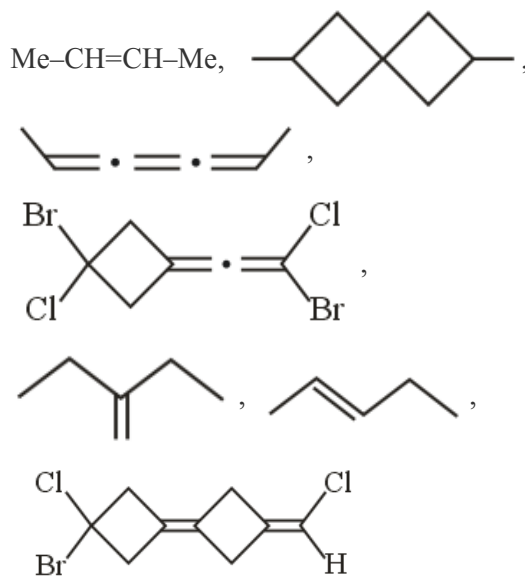
8. Among the following, the number of molecules which are optically active & possess C_2 axis of symmetry are :



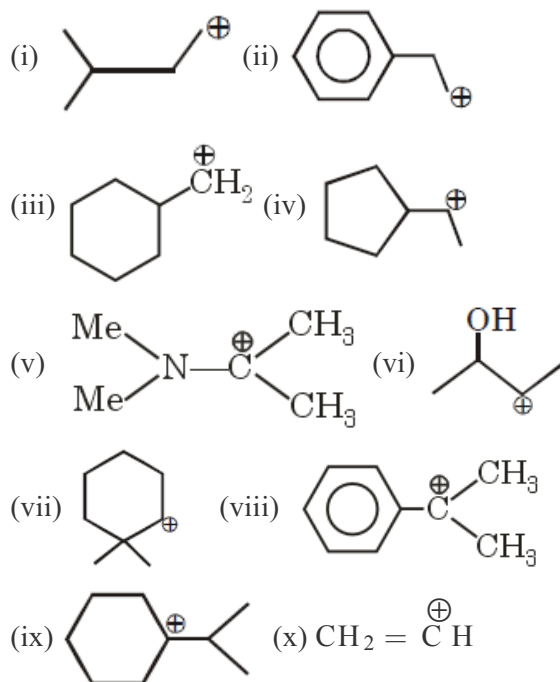
$CHCl_3$, 2,5-dimethylthiophene,



9. How many compound(s) can show geometrical isomerism ?



10. How many of the given carbocation can undergo rearrangement :



PART-3 : MATHEMATICS

SECTION - A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- If $a, b, c \in \mathbb{R}$, α and β are roots of equation $ax^2 + bx + c = 0$ ($a \neq 0, a \neq b \neq c$), then $(1 + \alpha + \alpha^2)(1 + \beta + \beta^2)$ will be
(A) 0 (B) Positive
(C) Negative (D) None of these
- If $a, b, c \in \mathbb{R}$ and quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) has no real roots then
(A) $c(a + b + c) < 0$
(B) $c(a + b + c) > 0$
(C) $a + b + c > 0$
(D) $c < 0$
- Number of real values of x for which $(x + 2)^2 + (x^2 + 5x + 6)^2 + (x^2 + 3x + 2)^2 = 0$ is
(A) 0 (B) 1
(C) 2 (D) 4
- The complete set of value of 'a' for which quadratic equation $x^2 + 2ax + a^2 - 1 = 0$ has at least one negative root is given by
(A) $(1, \infty)$
(B) $(-\infty, 1]$
(C) $(-1, \infty)$
(D) $(-\infty, -1]$

- Range of function f defined by $f(x) = \left[\frac{1}{\sin\{x\}} \right]$, (where $[.]$ and $\{.\}$ respectively denote the greatest integer function and the fractional part function) is
(A) $\mathbb{I}$, the set of integers
(B) $\mathbb{N}$, the set of natural numbers
(C) $\mathbb{W}$, the set of whole numbers
(D) $\{2, 3, 4, \dots\}$
- Domain of the function $f(x) = \sqrt{\sin x} - \sqrt{16 - x^2}$ is
(A) $x \in [0, \pi]$
(B) $x \in [-4, -\pi] \cup [0, \pi]$
(C) $x \in [0, 4]$
(D) $x \in [-4, 4]$
- Range of the function $f(x) = \sin^2 x - \sin x - 1$ is
(A) $[-1, 1]$
(B) $\left[-\frac{5}{4}, 1\right]$
(C) $\left[-\frac{3}{4}, 1\right]$
(D) $\left[-\frac{5}{4}, -1\right]$
- Let $f(x) = x^2 - 2x + 3$, $f : [0, 3] \rightarrow A$. The set 'A' for which $f(x)$ becomes onto -
(A) $[2, 6]$
(B) $[3, 6]$
(C) $[2, 3]$
(D) $[0, 3]$

9. Which of the following pairs is identical

(A) $\ln x^2$ and $2\ln x$

(B) $\ln x$ and $\frac{1}{\log_x e}$

(C) $\sec^2 x - \tan^2 x$ and 1

(D) $\frac{x}{x^2}$ and $\frac{1}{x}$

10. If domain of $f(x)$ is $(0, 1)$, then domain of $f(e^x)$ is -

(A) $(0, \infty)$

(B) $(1, \infty)$

(C) $(-\infty, 1)$

(D) $(-\infty, 0)$

11. Fundamental period of $f(x) = \sin^2 x + \cos^4 x$ is

(A) π

(B) 2π

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

12. Range of the function

$f(x) = \sin^{-1} x + \tan^{-1} x + \cos^{-1} x$ is

(A) $(0, \pi)$

(B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

(C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D) $[0, \pi]$

13. The value of

$$4 \left(\cos^{-1} \sin(-5) - 3 \tan^{-1} \left(\cot \left(\frac{-1}{4} \right) \right) \right)$$

is equal to

(A) 17

(B) 18

(C) 19

(D) 20

14. The number of values of x for which

$$\cos^{-1}(1-x) - 3\cos^{-1}x = \frac{\pi}{2}$$
 is

(A) 0

(B) 2

(C) 1

(D) 3

15. The value of

$$\tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) + \tan^{-1} \left(\frac{1}{13} \right) + \tan^{-1} \left(\frac{1}{21} \right) +$$

...infinite terms is $k\pi$, then the value of $8k$ is

(A) 1

(B) 2

(C) 3

(D) 5

16. Numbers of integral values of x for which $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \leq \frac{\pi}{3}$ is
 (A) 2 (B) 3
 (C) 0 (D) 1
17. If $1 \leq ||x-1|-2| < 2$ then the complete interval in which x^2 lies, is
 (A) $x^2 \in [0, 1) \cup (1, 9) \cup [16, 25)$
 (B) $x^2 \in [0, 1) \cup (10, 12) \cup [4, 5)$
 (C) $x^2 \in [0, 1) \cup (1, 9) \cup [12, 15)$
 (D) $x^2 \in (0, 1) \cup (30, 35)$
18. Polynomial $p(x)$ contains only terms of odd degree. If $p(x)$ is divided by $(x-3)$, the remainder is 6 and when $p(x)$ is divided by (x^2-9) then the remainder is $g(x)$. The value of $g(2)$ is
 (A) 2 (B) 4
 (C) 6 (D) 8
19. Let $f(x)$ be a monic polynomial function of degree 6 such that $f(1) = f(-1) = 5$, $f(2) = f(-2) = 8$ and $f(3) = f(-3) = 13$, then the value of $(f(\sqrt{2}) - f(\sqrt{3}))$ is
 (A) -2 (B) 0
 (C) 1 (D) 3

20. If $f(x)$ is a polynomial of degree 2 with leading coefficient 1 such that $|x^2 - x + 2 + f(x)| = |f(x)| = x^2 - x + 2 \forall x \in \mathbb{R}$ and $f(1) = 0$, then $f(3)$ is
 (A) 1 (B) 2
 (C) 3 (D) 4

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Let range of expression $\frac{x-2}{x^2-x+2}$ is $[\alpha, \beta] \forall x \in \mathbb{R}$, then value of $10|\alpha| + 7|\beta|$ is
2. If $x \geq 0, y \geq 0$, then area of the region bounded by $[x] + [y] = 50$ is
 (where $[.]$ represent greatest integral function)
3. Let $f(x) = \sqrt{-x^2 - 2x + n}$. If there are atleast 5 positive integer in the domain of $f(x)$ then minimum value of natural number n is

4. Let $f(x)$ be a function defined on $[0,1]$ such that
- $$f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 1-x & x \notin \mathbb{Q} \end{cases}$$
- and for all values of $x \in [0, 1]$, $f \circ f(x) = a + bx$, then the values of $10a + 20b$ is
5. If f be a greatest integer function and g be an absolute value function. The value of $(f \circ g)\left(\frac{-3}{2}\right) + (g \circ f)\left(\frac{4}{3}\right)$ is
6. The value of $\sin\left(4\tan^{-1}\frac{1}{3}\right) - \cos\left(2\tan^{-1}\frac{1}{7}\right)$ is
7. Let $f(x) = \cot^{-1}\left(\frac{1 + \sin 2x + \cos 2x}{1 + \sin 2x - \cos 2x}\right)$ and if $\sum_{x=1}^5 f(x) = 5p - q\pi$, where $p, q \in \mathbb{N}$, then $(p + q)$ is equal to
8. The number of solutions of the equation $\operatorname{sgn}(x) = \sin^{-1}x$ is
9. If the equation $x^2 - (2a + b)x + (2a^2 + b^2 - b + \frac{1}{2}) = 0$ has two real roots then $(2a + b)$ is equal to (where $a, b \in \mathbb{R}$)
10. If $\frac{2x}{2x^2 + 5x + 2} > \frac{1}{x + 1}$, then complete solution set is $(a, -1) \cup (b, c)$. The value of $5|a + 3b + 2c|$ is

Space for Rough Work

